

Course: B. Tech.

Branch: Civil Engineering/Civil & Environmental Engg

Subject Code & Name: BTCECS302/BTCVES302 & Mechanics of Solids

Semester: III

Max Marks: 60

Date: 04/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

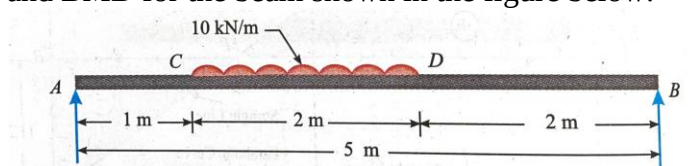
12

- A) Derive the relationship between modulus of elasticity and modulus of rigidity. **L2/CO1** **6**
- B) An Aluminum block 50 mm long placed over a Steel block 150 mm long. **L3/CO1** **6**
Both blocks have same cross-sectional area $A = 900 \text{ mm}^2$. The assembly carries a load P which deforms it by 0.30 mm. Determine the load P .
Take $E_{\text{aluminium}} = 7 \times 10^4 \text{ N/mm}^2$ and $E_{\text{steel}} = 2 \times 10^5 \text{ N/mm}^2$.
- C) A rectangular bar 300 mm long, 80 mm wide and 20 mm thick is loaded with **L3/CO1** **6**
an axial tensile load of 180 kN; together with a normal compressive force of 1800 kN on $80 \text{ mm} \times 300 \text{ mm}$ face and tensile force of 300 kN on $20 \text{ mm} \times 300 \text{ mm}$ face. Calculate change in length, width, thickness and volume.
Assume $E = 200 \text{ GPa}$ and $\mu = 0.3$

Q.2 Solve Any Two of the following.

12

- A) Draw the shear force diagram and bending moment diagram for a cantilever **L3/CO2** **6**
beam carrying a uniformly distributed load from its free end to mid-point.
- B) A horizontal beam AD, 15 m long carries a udl of 1.6 kN/m over whole span **L3/CO2** **6**
AD together with a concentrated load of 4 kN at the left end A. The beam is supported at a point B which is 1 m from A and at C which is 11 m from left end A. Draw SFD and BMD for this beam.
- C) Draw SFD and BMD for the beam shown in the figure below: **L3/CO2** **6**



Q. 3 Solve Any Two of the following.

12

- A) Define the neutral axis in the context of pure bending. Discuss how it relates **L2/CO3** **6**
to the distribution of bending stress.

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| B) | A 2.5 mm thick high strength copper strip 2.5 m long is bend into a circle and held with its ends just in contact. Find the maximum bending stress in the strip. How does this stress change if the thickness of the strip is increased? | L3/CO3 | 6 |
| C) | A hollow shaft is to transmit 400 kW at 160 r.p.m. If the shear stress is not to exceed 120 MPa and internal diameter is 0.6 of the external diameter, find the diameters of the shaft. | L3/CO3 | 6 |

Q.4 Solve Any Two of the following. **12**

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| A) | Define the terms "slenderness ratio" and "effective length" in the context of columns. | L1/CO4 | 6 |
| B) | A short length of tube 36 mm outside diameter and 24 mm inside diameter is tested under axial compression. In this test, tube failed at a load of 175 kN without buckling. When the tube of same section was tested as a strut with $L=1500$ mm both ends hinged, the Euler's load noted was 75 kN. Determine Rankine's constant. | L3/CO4 | 6 |
| C) | A hollow cast iron column whose outer diameter is 200 mm has a thickness of 20 mm. It is 4.5 meters long and is fixed at both the ends. Calculate the safe load by Rankine's formula using a factor of safety of 4. Calculate the slenderness ratio and the ratio of Euler's and Rankine's critical loads. For cast iron take $F_c = 550 \text{ N/mm}^2$ and Rankine's constant = $1/1600$ and $E = 80 \text{ GPa}$ | L3/CO4 | 6 |

Q. 5 Solve Any Two of the following. **12**

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|----|--|---------------|----------|
| A) | Explain the maximum principal stress theory. | L2/CO1 | 6 |
| B) | A plane element in a boiler is subjected to tensile stresses of 400 MPa on one plane and 150 MPa on the other at right angles to the former. Each of the above stresses is accompanied by a shear stress of 100 MPa such that when associated with the minor tensile stress tends to rotate the element in anticlockwise direction. Find <div style="margin-left: 40px;"> i. Principal stresses and their directions
 ii. Maximum shearing stresses and the directions of the plane on which they act </div> | L3/CO1 | 6 |
| C) | The stresses at a point of a machine components are 150 MPa and 50 MPa both tensile. Find the intensities of normal, shear and resultant stresses on a plane inclined at an angle of 55° with the axis of major tensile stress. Also find the magnitude of the maximum shear stresses in the components. | L3/CO1 | 6 |

***** End *****